

ResearchConnect SCSU

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Topics of Discussion

- Problem Statement & Features
- Team Contributions
- Design Specifications
- Implementation & Test Strategy
- ResearchConnect Demo

Problem Statement & Features

Problem, Motivation, and Solution

Problem

There isn't a centralized platform at SCSU where students can easily find research opportunities.

Students often rely on word of mouth, various websites, or direct faculty outreach, which makes it difficult to discover relevant opportunities.



Motivation

Many students are unaware of opportunities that align with their interests.

Professors may struggle to find motivated research assistants.

There is a need for greater visibility and accessibility of research opportunities for students.



Solution

ResearchConnect SCSU is a web-based platform that connects Computer Science and Data Science students to faculty-led research opportunities, as well as research and career-related resources at Southern.

What We've Built & Why It Matters

ResearchConnect SCSU is a web-based, AI-enhanced platform that centralizes research opportunities, benefitting both students and faculty

Deliverables	Features	Tools & Technologies	Impact
• User-friendly website that advertises research opportunities at SCSU • A chatbot that answers research questions • A database that stores and updates listings and user profiles • Full deployment for real-world access and hosting	• User authentication • User profiles • Content moderation • Research listings • Search and filter options • AI chatbot • Resource information	• Frontend - Python, Streamlit • Backend - Python • Database & Storage - Firebase • User Authentication - Microsoft Azure Active Directory • Generative AI - Google Cloud VertexAI, RAG • Collaboration - Git/GitHub • Deployment - Streamlit Community Cloud	• Students will have a centralized platform to efficiently explore research opportunities • Faculty will have an effective way to advertise research opportunities and connect with students

Team Contributions

Team Contributions

Tatiana Eng

Full-Stack Development and UI/UX Design

- Trello Board
- Figma Wireframe
- Class Diagram
- Listings Page with Student, Faculty, & Admin View
 - Browse Listings
 - Create Listing
 - View My Listings
- Resources Page

Orlando Marin

Full-Stack Development and Cloud Deployment

- Codebase Structure & Refactoring
- GitHub Repository Management
- Google Cloud & Firebase Projects
- Custom Theme & Logo Design
- Microsoft Azure OIDC Authentication
- Home Page
- Chatbot Page with VertexAI Integration
- Profile Page with Admin Dashboard

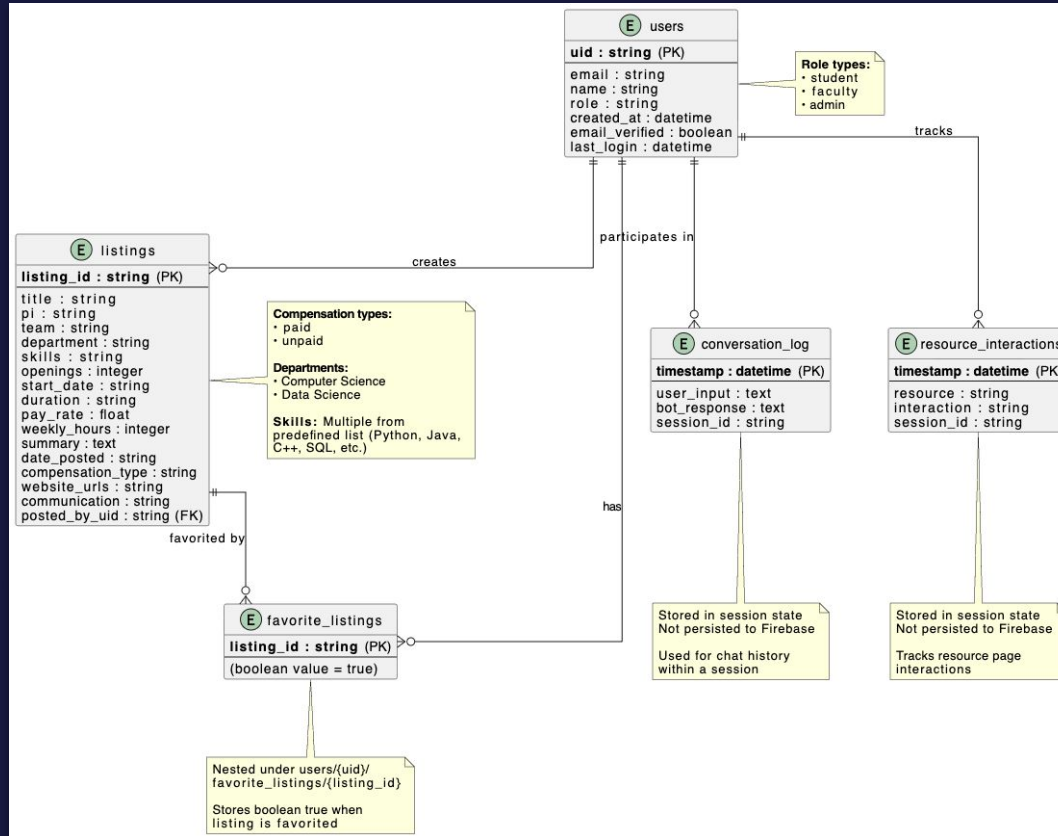
Sana Muneer

RAG Implementation and Quality Assurance

- User Stories
- Database Design & ER Diagram
- Initial Firebase Authentication Setup
- Retrieval Augmented Generation (RAG) AI Integration
- Testing & Quality Assurance
 - Unit Testing
 - Integration Testing
 - System Testing

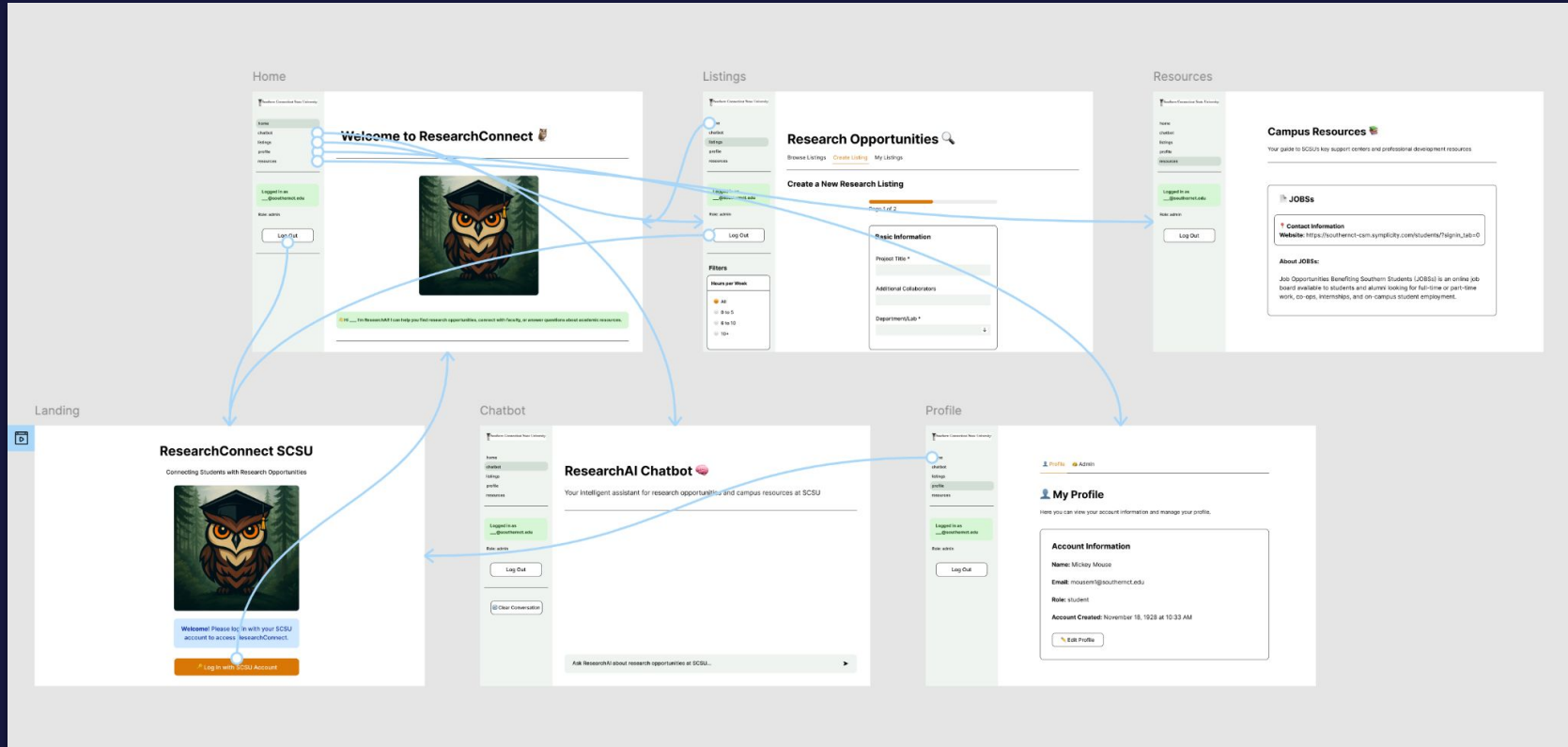
Design Specifications

Entity-Relationship (ER) Diagram



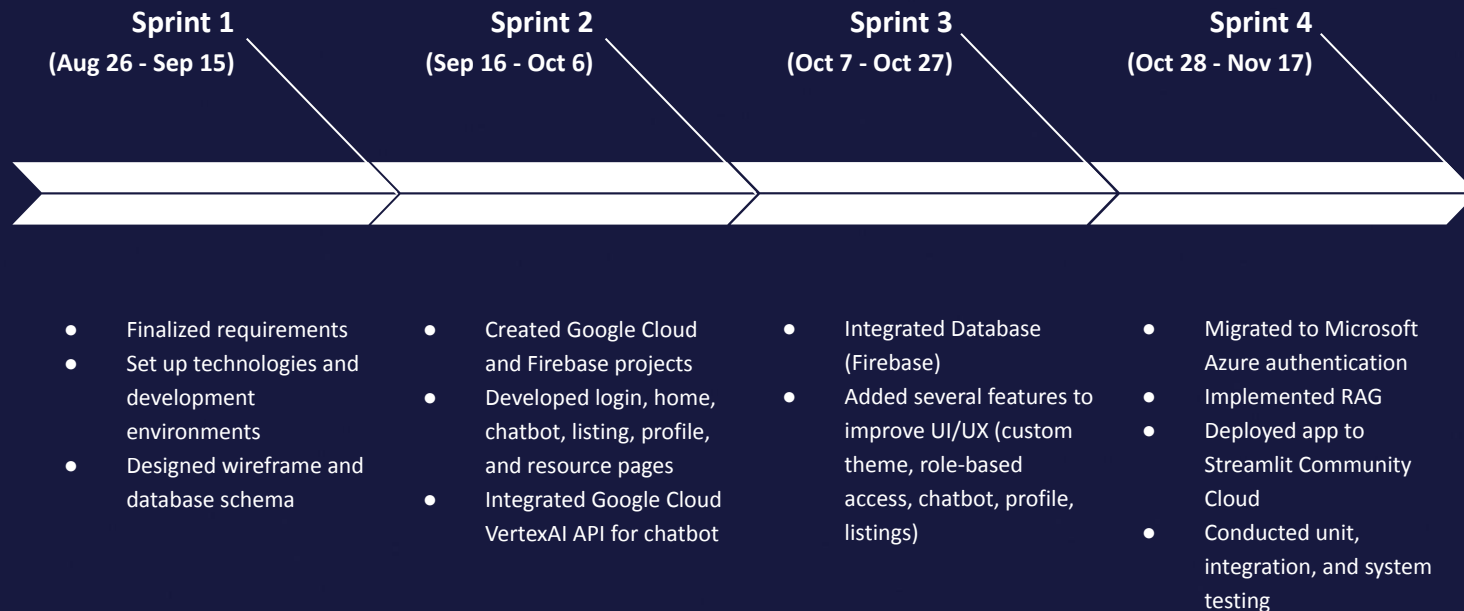


Figma Wireframe / UI Diagram



Implementation & Test Strategy

Project Timeline



Test Strategy

Unit Tests

Verify that functions work correctly in isolation



- Tested core functionality: authentication, chatbot prompts, RAG retrieval, listings (CRUD operations), and profile updates
- Detected and fixed logic errors to ensure component reliability
- Validated accurate, consistent results before integration

Integration Tests

Check that different modules or components of the system work together as expected



- Verified pages connect correctly with matching utils scripts
- Tested end-to-end flows (listings, chatbot RAG queries)
- Confirmed Firebase connectivity and data consistency
- Frequently pulled dev branch updates before continuing development

System Tests

Evaluate the system as a whole to ensure it meets the specified requirements and functions properly



- Conducted testing sessions with 3 students and 3 faculty performing real tasks: account creation, posting and browsing listings, and chatbot usage
- Captured feedback to identify improvement areas
- Verified responsiveness and role-based access before deployment

ResearchConnect Demo

Thank you!

References

- Figma Design Platform, <https://www.figma.com/>
- Firebase Documentation, <https://firebase.google.com/docs>
- Google Cloud Documentation, <https://cloud.google.com/docs>
- Microsoft Azure Documentation, <https://learn.microsoft.com/en-us/azure/?product=popular>
- Southern Connecticut State University Official Website, <https://www.southernct.edu/>
- Streamlit Documentation, <https://docs.streamlit.io/develop/api-reference>